

B. Statistical Tests

TABLE XVII: Post-hoc Wilcoxon tests with Holm correction for all factors. For each classifier (RF, XG, SVM, lgb, MLP, TabNet, TabFPN, TabTransformer) test results are shown for: (i) each pair of possible sample values; (ii) each pair of possible cluster values, and (iii) each pair of possible sample values, when isolating a specific cluster value

classifier	factor	level_a	level_b	raw_p	holm_p	significant
RF	n_clusters	5	10	0.00195	0.01172	True
RF	n_clusters	5	25	0.00195	0.00977	True
RF	n_clusters	5	50	0.00195	0.00781	True
RF	n_clusters	10	25	0.16016	0.16016	False
RF	n_clusters	10	50	0.00586	0.01758	True
RF	n_clusters	25	50	0.06445	0.12891	False
RF	n_samples	200	500	0.32227	0.32227	False
RF	n_samples	200	1000	0.00195	0.01172	True
RF	n_samples	200	2000	0.00195	0.00977	True
RF	n_samples	500	1000	0.00195	0.00781	True
RF	n_samples	500	2000	0.00195	0.00586	True
RF	n_samples	1000	2000	0.00195	0.00391	True
RF (n_clusters=5)	n_samples	200	500	0.01953	0.07812	False
RF (n_clusters=5)	n_samples	200	1000	0.00195	0.01172	True
RF (n_clusters=5)	n_samples	200	2000	0.00195	0.00977	True
RF (n_clusters=5)	n_samples	500	1000	0.04883	0.14648	False
RF (n_clusters=5)	n_samples	500	2000	0.04883	0.09766	False
RF (n_clusters=5)	n_samples	1000	2000	0.76953	0.76953	False
RF (n_clusters=10)	n_samples	200	500	0.02734	0.02734	True
RF (n_clusters=10)	n_samples	200	1000	0.00391	0.01562	True
RF (n_clusters=10)	n_samples	200	2000	0.00195	0.01172	True
RF (n_clusters=10)	n_samples	500	1000	0.01367	0.02734	True
RF (n_clusters=10)	n_samples	500	2000	0.00195	0.00977	True
RF (n_clusters=10)	n_samples	1000	2000	0.00977	0.02930	True
RF (n_clusters=25)	n_samples	200	500	0.00391	0.01172	True
RF (n_clusters=25)	n_samples	200	1000	0.76953	0.76953	False
RF (n_clusters=25)	n_samples	200	2000	0.03711	0.07422	False
RF (n_clusters=25)	n_samples	500	1000	0.00195	0.01172	True
RF (n_clusters=25)	n_samples	500	2000	0.00195	0.00977	True
RF (n_clusters=25)	n_samples	1000	2000	0.00195	0.00781	True
RF (n_clusters=50)	n_samples	200	500	0.16016	0.16016	False
RF (n_clusters=50)	n_samples	200	1000	0.00586	0.01172	True
RF (n_clusters=50)	n_samples	200	2000	0.00195	0.00977	True
RF (n_clusters=50)	n_samples	500	1000	0.00195	0.01172	True
RF (n_clusters=50)	n_samples	500	2000	0.00195	0.00586	True
RF (n_clusters=50)	n_samples	1000	2000	0.00195	0.00781	True
XG	n_clusters	5	10	0.00195	0.01172	True
XG	n_clusters	5	25	0.00195	0.00977	True
XG	n_clusters	5	50	0.00195	0.00781	True
XG	n_clusters	10	25	0.00195	0.00586	True
XG	n_clusters	10	50	0.00195	0.00391	True
XG	n_clusters	25	50	0.00195	0.00195	True
XG	n_samples	200	500	0.00195	0.01172	True
XG	n_samples	200	1000	0.00195	0.00977	True
XG	n_samples	200	2000	0.00195	0.00781	True
XG	n_samples	500	1000	0.00195	0.00586	True
XG	n_samples	500	2000	0.00195	0.00391	True
XG	n_samples	1000	2000	0.00195	0.00195	True
XG (n_clusters=5)	n_samples	200	500	0.23242	0.46484	False
XG (n_clusters=5)	n_samples	200	1000	0.02734	0.13672	False

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classifier	factor	level_a	level_b	raw_p	holm_p	significant
XG (n_clusters=5)	n_samples	200	2000	0.01367	0.08203	False
XG (n_clusters=5)	n_samples	500	1000	0.02734	0.10938	False
XG (n_clusters=5)	n_samples	500	2000	0.02734	0.08203	False
XG (n_clusters=5)	n_samples	1000	2000	0.49219	0.49219	False
XG (n_clusters=10)	n_samples	200	500	0.00586	0.01172	True
XG (n_clusters=10)	n_samples	200	1000	0.00195	0.01172	True
XG (n_clusters=10)	n_samples	200	2000	0.00195	0.00977	True
XG (n_clusters=10)	n_samples	500	1000	0.00195	0.00781	True
XG (n_clusters=10)	n_samples	500	2000	0.00195	0.00586	True
XG (n_clusters=10)	n_samples	1000	2000	0.00977	0.00977	True
XG (n_clusters=25)	n_samples	200	500	0.00391	0.00781	True
XG (n_clusters=25)	n_samples	200	1000	0.00977	0.00977	True
XG (n_clusters=25)	n_samples	200	2000	0.00195	0.00977	True
XG (n_clusters=25)	n_samples	500	1000	0.00195	0.01172	True
XG (n_clusters=25)	n_samples	500	2000	0.00195	0.00586	True
XG (n_clusters=25)	n_samples	1000	2000	0.00195	0.00781	True
XG (n_clusters=50)	n_samples	200	500	0.00391	0.00391	True
XG (n_clusters=50)	n_samples	200	1000	0.00195	0.01172	True
XG (n_clusters=50)	n_samples	200	2000	0.00195	0.00977	True
XG (n_clusters=50)	n_samples	500	1000	0.00195	0.00781	True
XG (n_clusters=50)	n_samples	500	2000	0.00195	0.00586	True
XG (n_clusters=50)	n_samples	1000	2000	0.00195	0.00391	True
SVM	n_clusters	5	10	0.00195	0.01172	True
SVM	n_clusters	5	25	0.00195	0.00977	True
SVM	n_clusters	5	50	0.00195	0.00781	True
SVM	n_clusters	10	25	0.00195	0.00586	True
SVM	n_clusters	10	50	0.00195	0.00391	True
SVM	n_clusters	25	50	0.00195	0.00195	True
SVM	n_samples	200	500	0.00195	0.01172	True
SVM	n_samples	200	1000	0.00195	0.00977	True
SVM	n_samples	200	2000	0.00195	0.00781	True
SVM	n_samples	500	1000	0.00195	0.00586	True
SVM	n_samples	500	2000	0.00195	0.00391	True
SVM	n_samples	1000	2000	0.00195	0.00195	True
SVM (n_clusters=5)	n_samples	200	500	0.32227	0.64453	False
SVM (n_clusters=5)	n_samples	200	1000	0.02734	0.10938	False
SVM (n_clusters=5)	n_samples	200	2000	0.55664	0.55664	False
SVM (n_clusters=5)	n_samples	500	1000	0.01953	0.09766	False
SVM (n_clusters=5)	n_samples	500	2000	0.16016	0.48047	False
SVM (n_clusters=5)	n_samples	1000	2000	0.00195	0.01172	True
SVM (n_clusters=10)	n_samples	200	500	0.06445	0.25781	False
SVM (n_clusters=10)	n_samples	200	1000	0.13086	0.39258	False
SVM (n_clusters=10)	n_samples	200	2000	0.04883	0.29297	False
SVM (n_clusters=10)	n_samples	500	1000	0.55664	0.55664	False
SVM (n_clusters=10)	n_samples	500	2000	0.27539	0.55078	False
SVM (n_clusters=10)	n_samples	1000	2000	0.04883	0.24414	False
SVM (n_clusters=25)	n_samples	200	500	0.00195	0.01172	True
SVM (n_clusters=25)	n_samples	200	1000	0.00195	0.00977	True
SVM (n_clusters=25)	n_samples	200	2000	0.00195	0.00781	True
SVM (n_clusters=25)	n_samples	500	1000	0.00195	0.00586	True
SVM (n_clusters=25)	n_samples	500	2000	0.00195	0.00391	True
SVM (n_clusters=25)	n_samples	1000	2000	0.00195	0.00195	True
SVM (n_clusters=50)	n_samples	200	500	0.00195	0.01172	True
SVM (n_clusters=50)	n_samples	200	1000	0.00195	0.00977	True
SVM (n_clusters=50)	n_samples	200	2000	0.00195	0.00781	True

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classifier	factor	level_a	level_b	raw_p	holm_p	significant
SVM (n_clusters=50)	n_samples	500	1000	0.00195	0.00586	True
SVM (n_clusters=50)	n_samples	500	2000	0.00195	0.00391	True
SVM (n_clusters=50)	n_samples	1000	2000	0.00195	0.00195	True
lgb	n_clusters	5	10	0.00195	0.01172	True
lgb	n_clusters	5	25	0.00195	0.00977	True
lgb	n_clusters	5	50	0.00195	0.00781	True
lgb	n_clusters	10	25	0.00195	0.00586	True
lgb	n_clusters	10	50	0.00195	0.00391	True
lgb	n_clusters	25	50	0.00195	0.00195	True
lgb	n_samples	200	500	0.00195	0.01172	True
lgb	n_samples	200	1000	0.00195	0.00977	True
lgb	n_samples	200	2000	0.00195	0.00781	True
lgb	n_samples	500	1000	0.00195	0.00586	True
lgb	n_samples	500	2000	0.00195	0.00391	True
lgb	n_samples	1000	2000	0.00195	0.00195	True
lgb (n_clusters=5)	n_samples	200	500	0.01953	0.03906	True
lgb (n_clusters=5)	n_samples	200	1000	0.00195	0.01172	True
lgb (n_clusters=5)	n_samples	200	2000	0.00195	0.00977	True
lgb (n_clusters=5)	n_samples	500	1000	0.00977	0.02930	True
lgb (n_clusters=5)	n_samples	500	2000	0.00391	0.01562	True
lgb (n_clusters=5)	n_samples	1000	2000	0.62500	0.62500	False
lgb (n_clusters=10)	n_samples	200	500	0.03711	0.03711	True
lgb (n_clusters=10)	n_samples	200	1000	0.00391	0.00781	True
lgb (n_clusters=10)	n_samples	200	2000	0.00195	0.00977	True
lgb (n_clusters=10)	n_samples	500	1000	0.00195	0.01172	True
lgb (n_clusters=10)	n_samples	500	2000	0.00195	0.00586	True
lgb (n_clusters=10)	n_samples	1000	2000	0.00195	0.00781	True
lgb (n_clusters=25)	n_samples	200	500	0.84570	0.84570	False
lgb (n_clusters=25)	n_samples	200	1000	0.00195	0.01172	True
lgb (n_clusters=25)	n_samples	200	2000	0.00195	0.00977	True
lgb (n_clusters=25)	n_samples	500	1000	0.00195	0.00781	True
lgb (n_clusters=25)	n_samples	500	2000	0.00195	0.00586	True
lgb (n_clusters=25)	n_samples	1000	2000	0.00195	0.00391	True
lgb (n_clusters=50)	n_samples	200	500	0.00195	0.01172	True
lgb (n_clusters=50)	n_samples	200	1000	0.00195	0.00977	True
lgb (n_clusters=50)	n_samples	200	2000	0.00195	0.00781	True
lgb (n_clusters=50)	n_samples	500	1000	0.00195	0.00586	True
lgb (n_clusters=50)	n_samples	500	2000	0.00195	0.00391	True
lgb (n_clusters=50)	n_samples	1000	2000	0.00195	0.00195	True
MLP	n_clusters	5	10	0.00195	0.01172	True
MLP	n_clusters	5	25	0.00195	0.00977	True
MLP	n_clusters	5	50	0.00195	0.00781	True
MLP	n_clusters	10	25	0.00195	0.00586	True
MLP	n_clusters	10	50	0.00195	0.00391	True
MLP	n_clusters	25	50	0.00195	0.00195	True
MLP	n_samples	200	500	0.00195	0.01172	True
MLP	n_samples	200	1000	0.00195	0.00977	True
MLP	n_samples	200	2000	0.00195	0.00781	True
MLP	n_samples	500	1000	0.00195	0.00586	True
MLP	n_samples	500	2000	0.00195	0.00391	True
MLP	n_samples	1000	2000	0.00195	0.00195	True
MLP (n_clusters=5)	n_samples	200	500	0.84570	0.84570	False
MLP (n_clusters=5)	n_samples	200	1000	0.43164	1.00000	False
MLP (n_clusters=5)	n_samples	200	2000	0.84570	1.00000	False
MLP (n_clusters=5)	n_samples	500	1000	0.49219	1.00000	False

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classifier	factor	level_a	level_b	raw_p	holm_p	significant
MLP (n_clusters=5)	n_samples	500	2000	0.49219	1.00000	False
MLP (n_clusters=5)	n_samples	1000	2000	0.19336	1.00000	False
MLP (n_clusters=10)	n_samples	200	500	0.00977	0.02930	True
MLP (n_clusters=10)	n_samples	200	1000	0.00195	0.01172	True
MLP (n_clusters=10)	n_samples	200	2000	0.00195	0.00977	True
MLP (n_clusters=10)	n_samples	500	1000	0.01953	0.03906	True
MLP (n_clusters=10)	n_samples	500	2000	0.00195	0.00781	True
MLP (n_clusters=10)	n_samples	1000	2000	0.04883	0.04883	True
MLP (n_clusters=25)	n_samples	200	500	0.49219	0.49219	False
MLP (n_clusters=25)	n_samples	200	1000	0.00195	0.01172	True
MLP (n_clusters=25)	n_samples	200	2000	0.00195	0.00977	True
MLP (n_clusters=25)	n_samples	500	1000	0.00195	0.00781	True
MLP (n_clusters=25)	n_samples	500	2000	0.00195	0.00586	True
MLP (n_clusters=25)	n_samples	1000	2000	0.00195	0.00391	True
MLP (n_clusters=50)	n_samples	200	500	0.00195	0.01172	True
MLP (n_clusters=50)	n_samples	200	1000	0.00195	0.00977	True
MLP (n_clusters=50)	n_samples	200	2000	0.00195	0.00781	True
MLP (n_clusters=50)	n_samples	500	1000	0.00195	0.00586	True
MLP (n_clusters=50)	n_samples	500	2000	0.00195	0.00391	True
MLP (n_clusters=50)	n_samples	1000	2000	0.00195	0.00195	True
TabNet	n_clusters	5	10	0.00391	0.01562	True
TabNet	n_clusters	5	25	0.03711	0.07422	False
TabNet	n_clusters	5	50	0.00195	0.01172	True
TabNet	n_clusters	10	25	0.49219	0.49219	False
TabNet	n_clusters	10	50	0.00195	0.00977	True
TabNet	n_clusters	25	50	0.00586	0.01758	True
TabNet	n_samples	200	500	0.02734	0.02734	True
TabNet	n_samples	200	1000	0.00195	0.01172	True
TabNet	n_samples	200	2000	0.00195	0.00977	True
TabNet	n_samples	500	1000	0.00195	0.00781	True
TabNet	n_samples	500	2000	0.00195	0.00586	True
TabNet	n_samples	1000	2000	0.00195	0.00391	True
TabNet (n_clusters=5)	n_samples	200	500	0.27539	0.82617	False
TabNet (n_clusters=5)	n_samples	200	1000	0.08398	0.41992	False
TabNet (n_clusters=5)	n_samples	200	2000	0.06445	0.38672	False
TabNet (n_clusters=5)	n_samples	500	1000	0.49219	0.49219	False
TabNet (n_clusters=5)	n_samples	500	2000	0.16016	0.64062	False
TabNet (n_clusters=5)	n_samples	1000	2000	0.27539	0.55078	False
TabNet (n_clusters=10)	n_samples	200	500	0.00977	0.02930	True
TabNet (n_clusters=10)	n_samples	200	1000	0.00195	0.01172	True
TabNet (n_clusters=10)	n_samples	200	2000	0.00195	0.00977	True
TabNet (n_clusters=10)	n_samples	500	1000	0.00977	0.01953	True
TabNet (n_clusters=10)	n_samples	500	2000	0.00195	0.00781	True
TabNet (n_clusters=10)	n_samples	1000	2000	0.23242	0.23242	False
TabNet (n_clusters=25)	n_samples	200	500	0.00391	0.01562	True
TabNet (n_clusters=25)	n_samples	200	1000	0.62500	0.62500	False
TabNet (n_clusters=25)	n_samples	200	2000	0.10547	0.21094	False
TabNet (n_clusters=25)	n_samples	500	1000	0.00195	0.01172	True
TabNet (n_clusters=25)	n_samples	500	2000	0.00195	0.00977	True
TabNet (n_clusters=25)	n_samples	1000	2000	0.01953	0.05859	False
TabNet (n_clusters=50)	n_samples	200	500	0.04883	0.04883	True
TabNet (n_clusters=50)	n_samples	200	1000	0.00195	0.01172	True
TabNet (n_clusters=50)	n_samples	200	2000	0.00195	0.00977	True
TabNet (n_clusters=50)	n_samples	500	1000	0.00195	0.00781	True
TabNet (n_clusters=50)	n_samples	500	2000	0.00195	0.00586	True

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classifier	factor	level_a	level_b	raw_p	holm_p	significant
TabNet (n_clusters=50)	n_samples	1000	2000	0.00195	0.00391	True
TabPFN	n_clusters	5	10	0.00195	0.01172	True
TabPFN	n_clusters	5	25	0.00586	0.02930	True
TabPFN	n_clusters	5	50	0.00977	0.03906	True
TabPFN	n_clusters	10	25	0.43164	1.00000	False
TabPFN	n_clusters	10	50	0.49219	0.98438	False
TabPFN	n_clusters	25	50	1.00000	1.00000	False
TabPFN	n_samples	200	500	0.00977	0.01953	True
TabPFN	n_samples	200	1000	0.08398	0.08398	False
TabPFN	n_samples	200	2000	0.00195	0.00977	True
TabPFN	n_samples	500	1000	0.00195	0.01172	True
TabPFN	n_samples	500	2000	0.00195	0.00586	True
TabPFN	n_samples	1000	2000	0.00195	0.00781	True
TabPFN (n_clusters=5)	n_samples	200	500	0.01953	0.05859	False
TabPFN (n_clusters=5)	n_samples	200	1000	0.00195	0.01172	True
TabPFN (n_clusters=5)	n_samples	200	2000	0.00195	0.00977	True
TabPFN (n_clusters=5)	n_samples	500	1000	0.01367	0.05469	False
TabPFN (n_clusters=5)	n_samples	500	2000	0.01953	0.03906	True
TabPFN (n_clusters=5)	n_samples	1000	2000	0.84570	0.84570	False
TabPFN (n_clusters=10)	n_samples	200	500	0.10547	0.10547	False
TabPFN (n_clusters=10)	n_samples	200	1000	0.01953	0.05859	False
TabPFN (n_clusters=10)	n_samples	200	2000	0.00195	0.01172	True
TabPFN (n_clusters=10)	n_samples	500	1000	0.02734	0.05469	False
TabPFN (n_clusters=10)	n_samples	500	2000	0.00391	0.01953	True
TabPFN (n_clusters=10)	n_samples	1000	2000	0.00977	0.03906	True
TabPFN (n_clusters=25)	n_samples	200	500	0.00391	0.01172	True
TabPFN (n_clusters=25)	n_samples	200	1000	0.08398	0.16797	False
TabPFN (n_clusters=25)	n_samples	200	2000	0.76953	0.76953	False
TabPFN (n_clusters=25)	n_samples	500	1000	0.00195	0.01172	True
TabPFN (n_clusters=25)	n_samples	500	2000	0.00195	0.00977	True
TabPFN (n_clusters=25)	n_samples	1000	2000	0.00195	0.00781	True
TabPFN (n_clusters=50)	n_samples	200	500	0.00195	0.01172	True
TabPFN (n_clusters=50)	n_samples	200	1000	0.02734	0.05469	False
TabPFN (n_clusters=50)	n_samples	200	2000	0.27539	0.27539	False
TabPFN (n_clusters=50)	n_samples	500	1000	0.00195	0.00977	True
TabPFN (n_clusters=50)	n_samples	500	2000	0.00195	0.00586	True
TabPFN (n_clusters=50)	n_samples	1000	2000	0.00195	0.00781	True
TabTransformer	n_clusters	5	10	0.00195	0.01172	True
TabTransformer	n_clusters	5	25	0.00195	0.00977	True
TabTransformer	n_clusters	5	50	0.00195	0.00781	True
TabTransformer	n_clusters	10	25	0.00195	0.00586	True
TabTransformer	n_clusters	10	50	0.00195	0.00391	True
TabTransformer	n_clusters	25	50	0.00195	0.00195	True
TabTransformer	n_samples	200	500	0.00195	0.01172	True
TabTransformer	n_samples	200	1000	0.00195	0.00977	True
TabTransformer	n_samples	200	2000	0.00195	0.00781	True
TabTransformer	n_samples	500	1000	0.00195	0.00586	True
TabTransformer	n_samples	500	2000	0.00195	0.00391	True
TabTransformer	n_samples	1000	2000	0.00195	0.00195	True
TabTransformer (n_clusters=5)	n_samples	200	500	0.92188	1.00000	False
TabTransformer (n_clusters=5)	n_samples	200	1000	0.27539	1.00000	False
TabTransformer (n_clusters=5)	n_samples	200	2000	1.00000	1.00000	False
TabTransformer (n_clusters=5)	n_samples	500	1000	0.13086	0.65430	False
TabTransformer (n_clusters=5)	n_samples	500	2000	0.92188	1.00000	False
TabTransformer (n_clusters=5)	n_samples	1000	2000	0.00195	0.01172	True

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classifier	factor	level_a	level_b	raw_p	holm_p	significant
TabTransformer (n_clusters=10)	n_samples	200	500	0.00977	0.01953	True
TabTransformer (n_clusters=10)	n_samples	200	1000	0.00195	0.01172	True
TabTransformer (n_clusters=10)	n_samples	200	2000	0.00195	0.00977	True
TabTransformer (n_clusters=10)	n_samples	500	1000	0.00391	0.01172	True
TabTransformer (n_clusters=10)	n_samples	500	2000	0.00195	0.00781	True
TabTransformer (n_clusters=10)	n_samples	500	2000	0.00195	0.00781	True
TabTransformer (n_clusters=10)	n_samples	1000	2000	0.00977	0.00977	True
TabTransformer (n_clusters=10)	n_samples	500	2000	0.00195	0.00781	True
TabTransformer (n_clusters=10)	n_samples	500	2000	0.00195	0.00781	True
TabTransformer (n_clusters=10)	n_samples	1000	2000	0.00977	0.00977	True
TabTransformer (n_clusters=25)	n_samples	200	500	0.92188	0.92188	False
TabTransformer (n_clusters=25)	n_samples	200	1000	0.00195	0.01172	True
TabTransformer (n_clusters=25)	n_samples	200	2000	0.00195	0.00977	True
TabTransformer (n_clusters=25)	n_samples	500	1000	0.00195	0.00781	True
TabTransformer (n_clusters=25)	n_samples	500	2000	0.00195	0.00586	True
TabTransformer (n_clusters=25)	n_samples	1000	2000	0.00195	0.00391	True
TabTransformer (n_clusters=50)	n_samples	200	500	0.00195	0.01172	True
TabTransformer (n_clusters=50)	n_samples	200	1000	0.00195	0.00977	True
TabTransformer (n_clusters=50)	n_samples	200	2000	0.00195	0.00781	True
TabTransformer (n_clusters=50)	n_samples	500	1000	0.00195	0.00586	True
TabTransformer (n_clusters=50)	n_samples	500	2000	0.00195	0.00391	True
TabTransformer (n_clusters=50)	n_samples	1000	2000	0.00195	0.00195	True